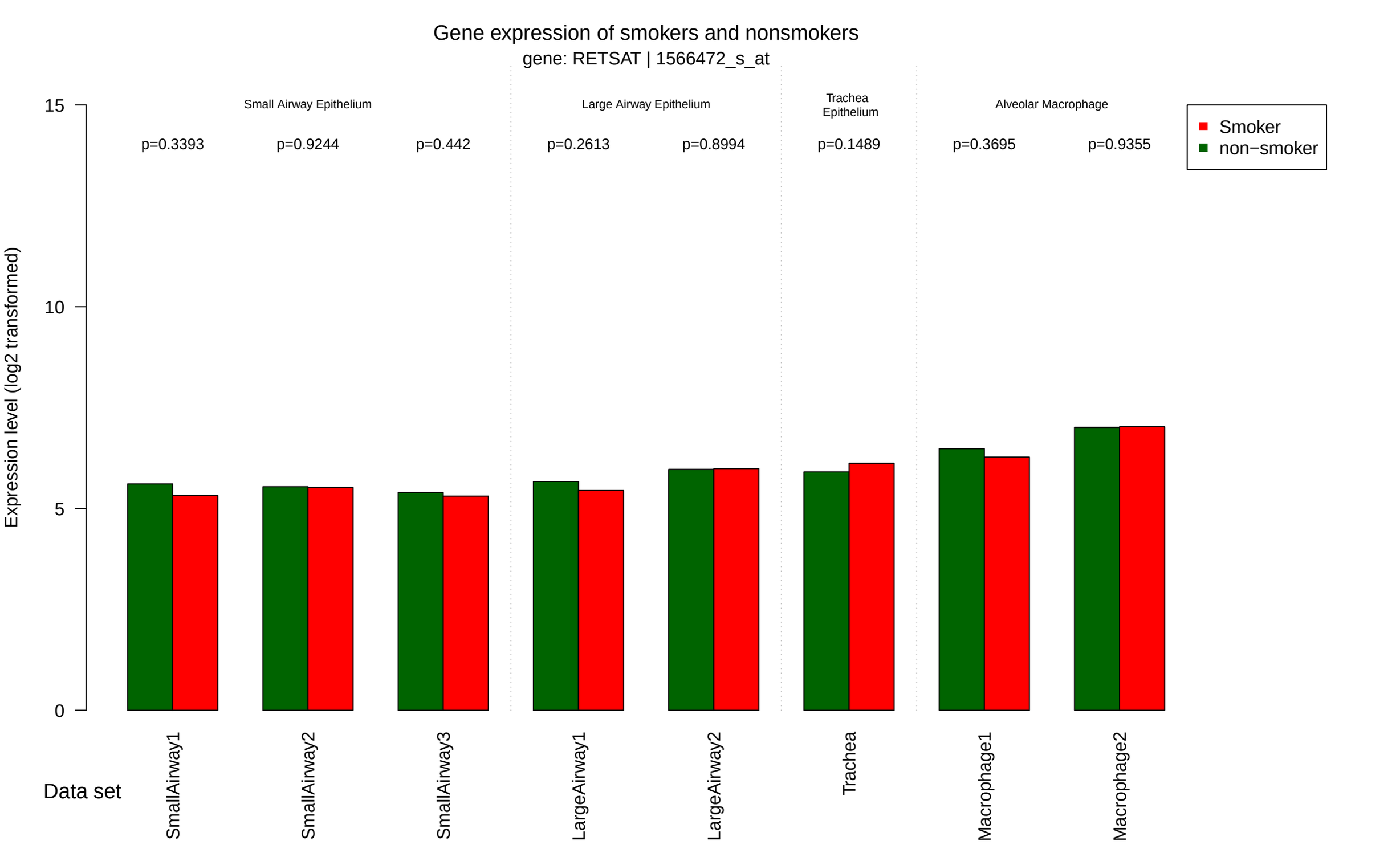
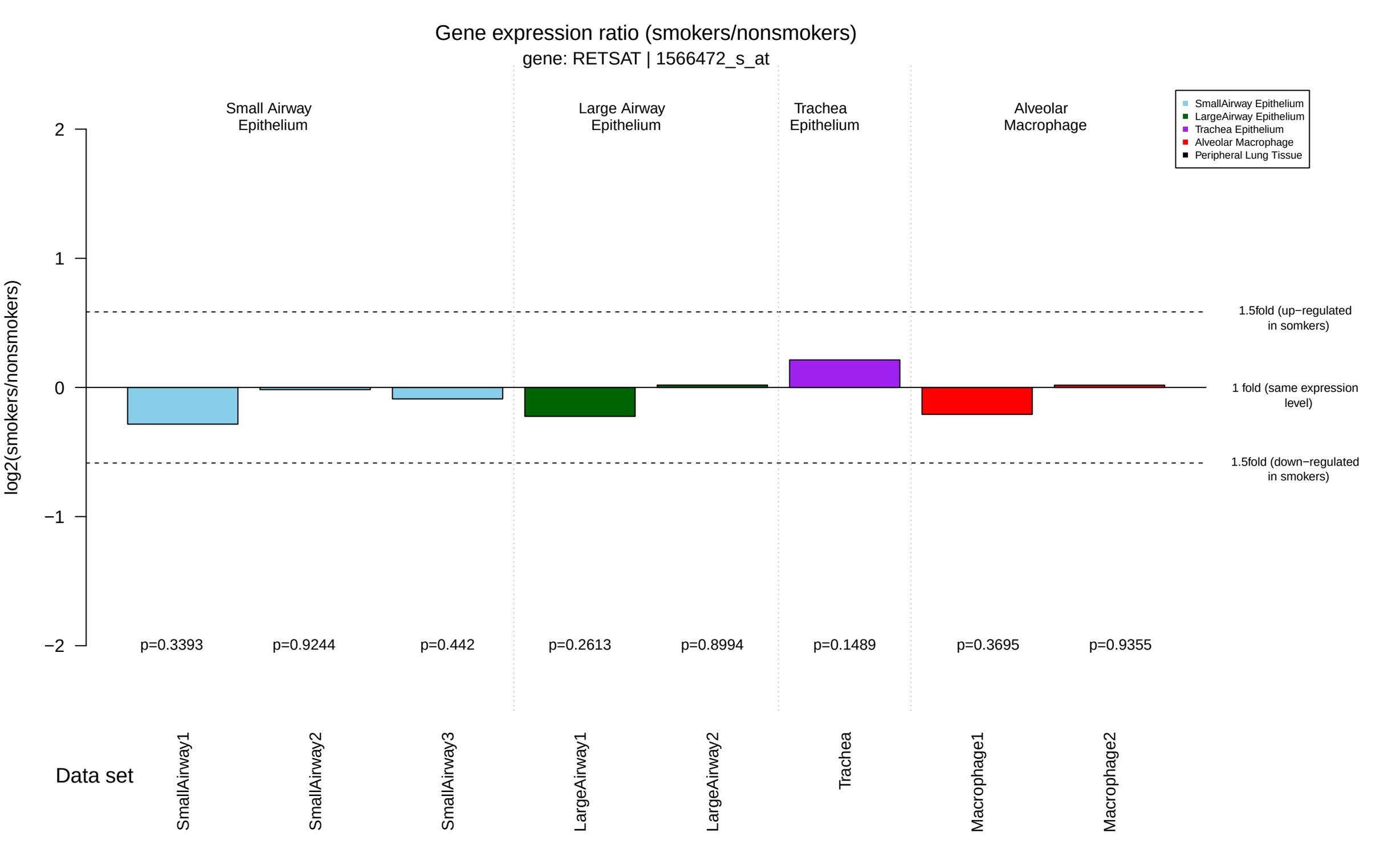
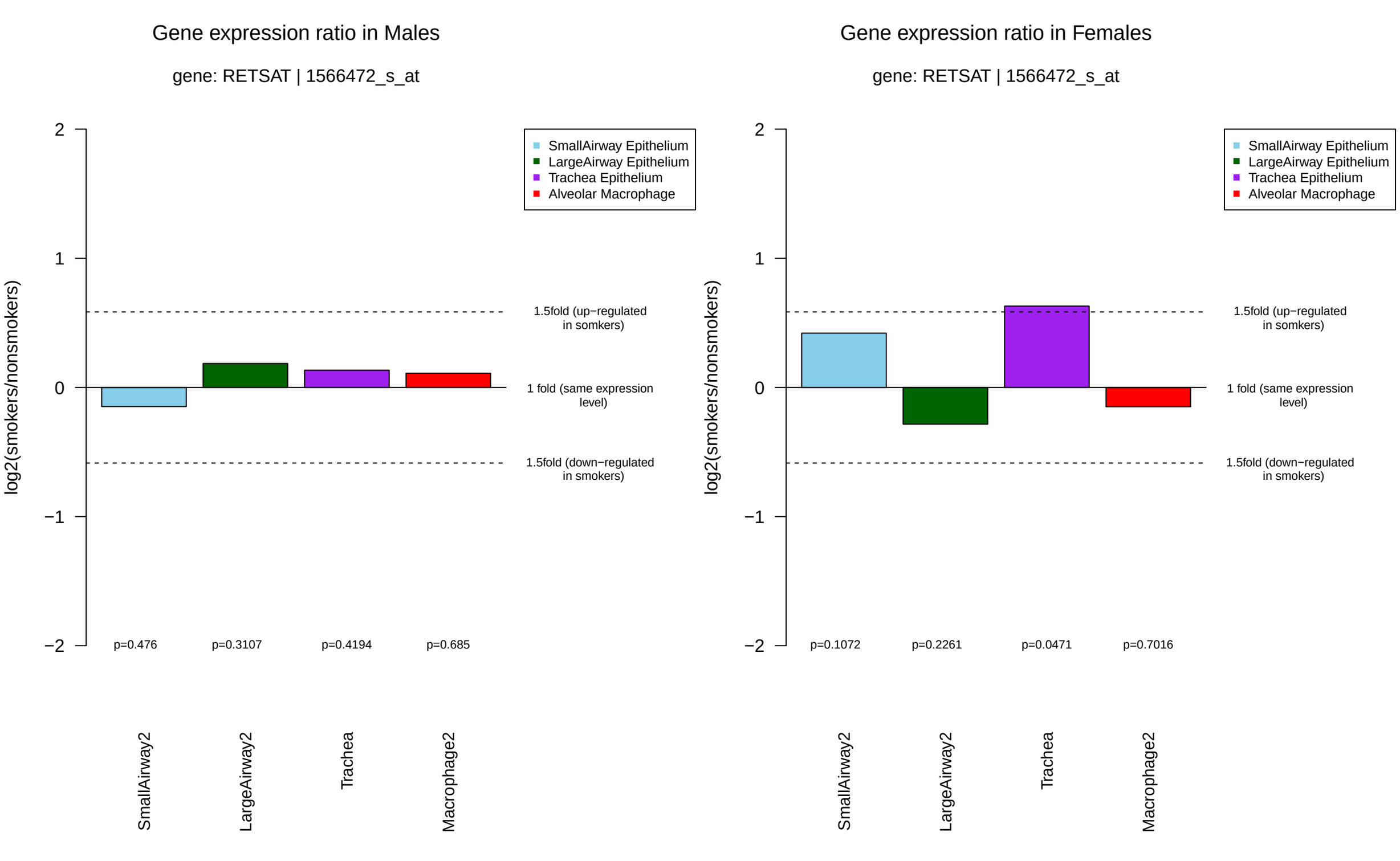
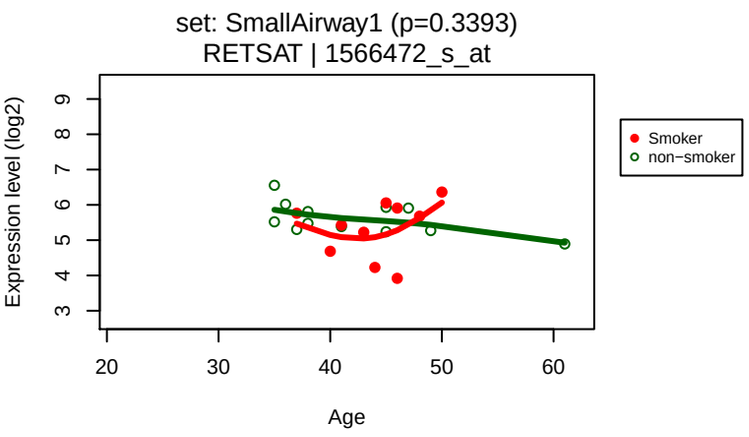
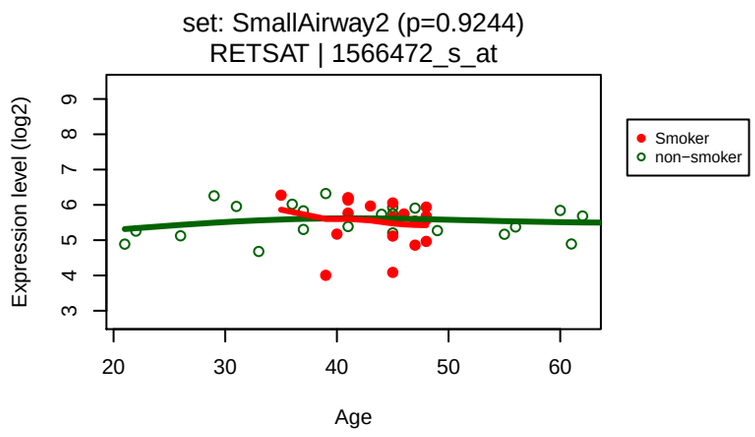
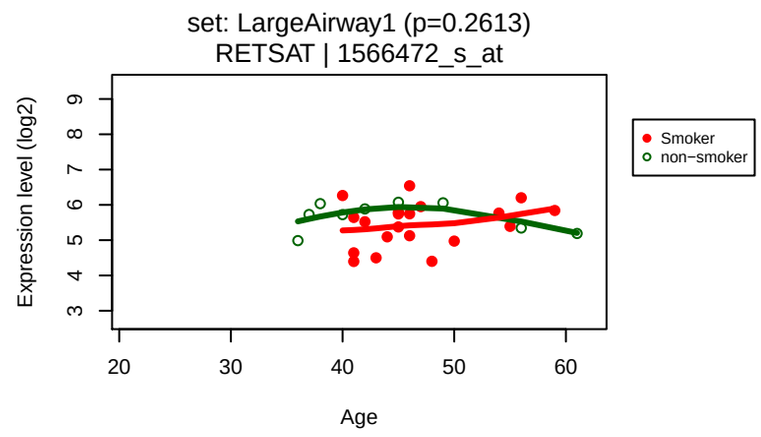

Gene expression of smokers and nonsmokers
gene: RETSAT | 1566472_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3393
p=0.9244
p=0.442
p=0.2613
p=0.8994
p=0.1489
p=0.3695
p=0.9355
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

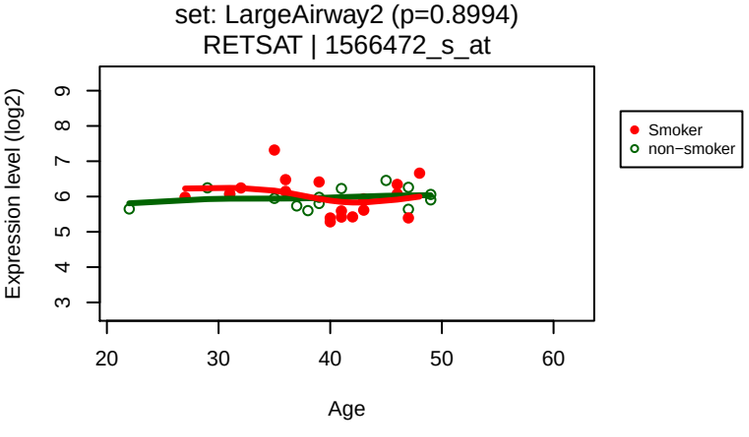

Gene expression ratio (smokers/nonsmokers)
gene: RETSAT | 1566472_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3393
p=0.9244
p=0.442
p=0.2613
p=0.8994
p=0.1489
p=0.3695
p=0.9355
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

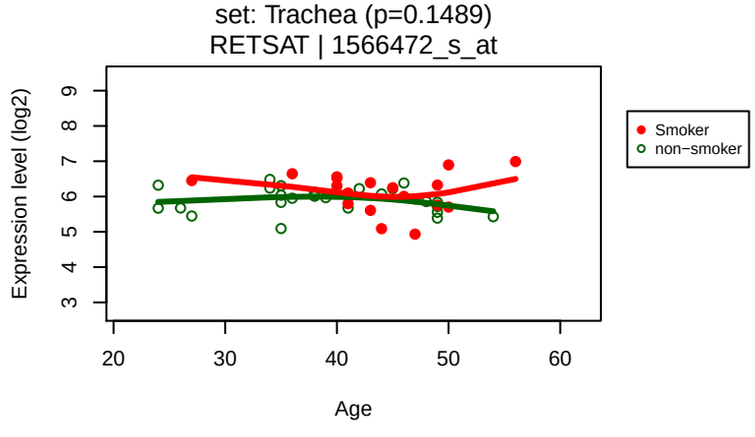

Gene expression ratio in Males
gene: RETSAT | 1566472_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.476
p=0.3107
p=0.4194
p=0.685
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RETSAT | 1566472_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1072
p=0.2261
p=0.0471
p=0.7016
SmallAirway2
LargeAirway2
Trachea
Macrophage2

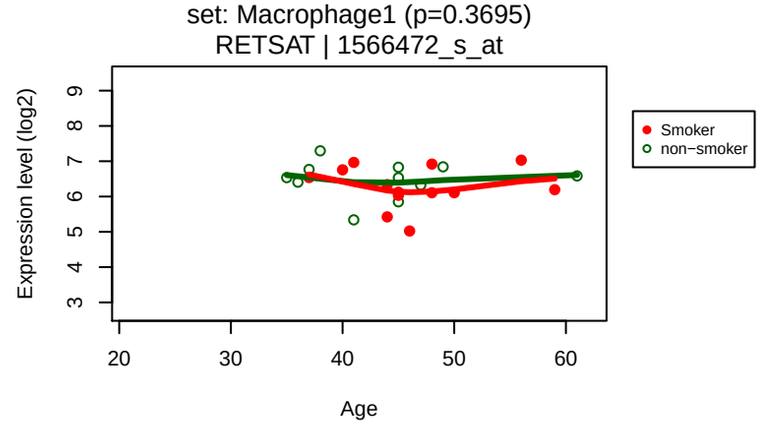

set: SmallAirway1 (p=0.3393)
RETSAT | 1566472_s_at
Expression level (log2)
Age
Smoker
non−smoker

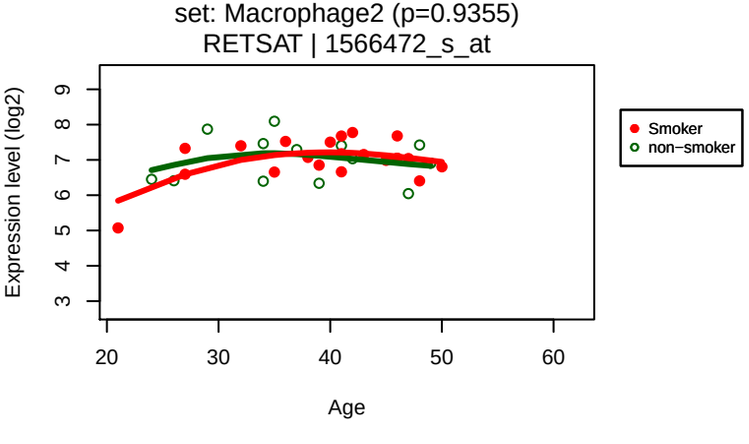

set: SmallAirway2 (p=0.9244)
RETSAT | 1566472_s_at
Expression level (log2)
Age
Smoker
non−smoker

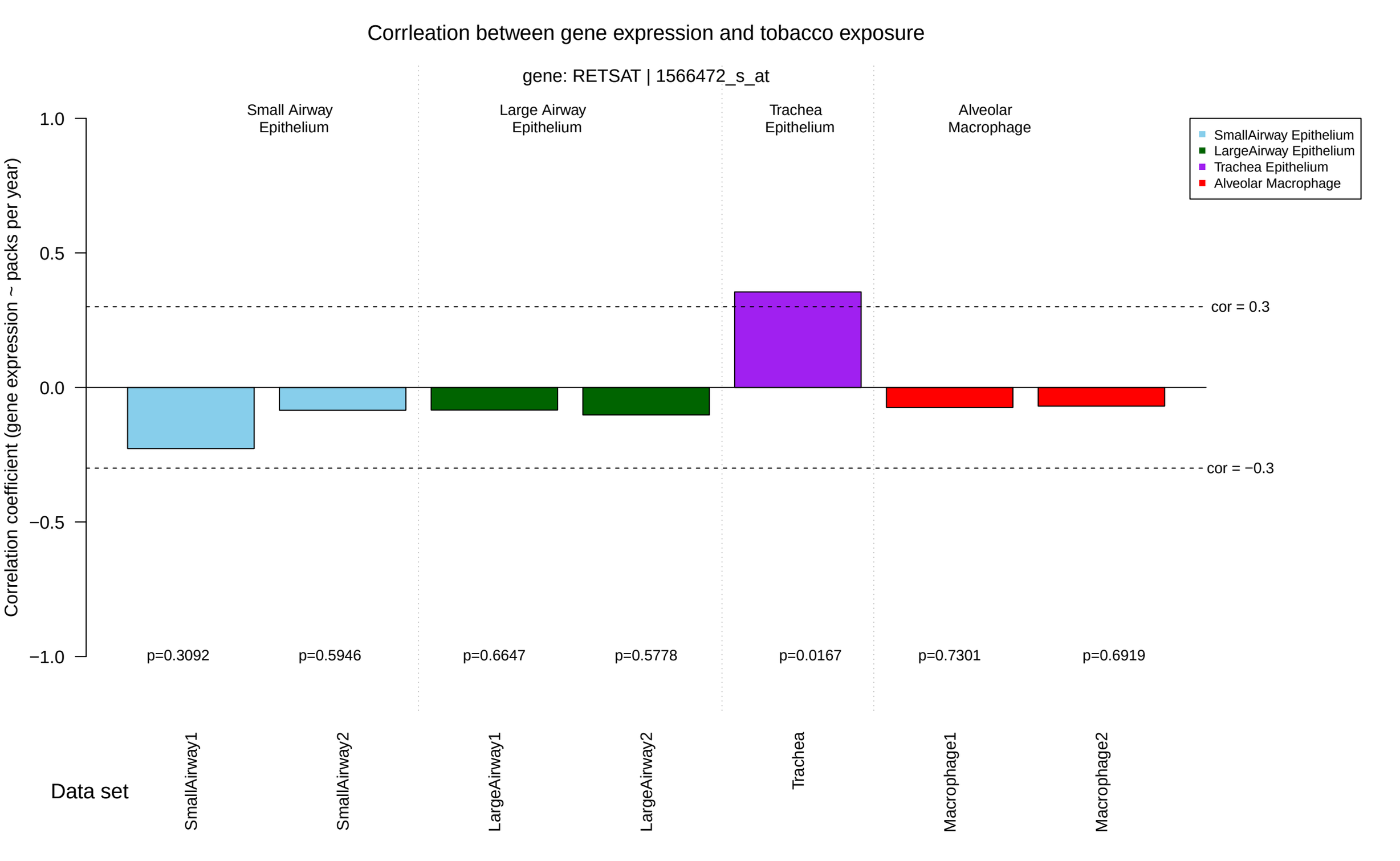

set: LargeAirway1 (p=0.2613)
RETSAT | 1566472_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8994)
RETSAT | 1566472_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1489)
RETSAT | 1566472_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3695)
RETSAT | 1566472_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9355)
RETSAT | 1566472_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RETSAT | 1566472_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3092
p=0.5946
p=0.6647
p=0.5778
p=0.0167
p=0.7301
p=0.6919
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2